Context-Environment Gaps: Benchmarking Tool-Using Agent Safety in Benign Tasks

Abstract

Tool-using agents can cause serious harm during everyday tasks even when users have benign intentions and no adversary is present. Existing safety evaluations in such settings use different failure categories and test conditions, obscuring the common structure behind seemingly distinct failures. We therefore propose context-environment gaps as a unifying explanation for these failures. Context is fully accessible to agents and is often treated as trustworthy, but it is not necessarily a reliable basis for safe action and requires validation against the actual environment. However, the environment is only partially observable through tool calls, so agents may not obtain the evidence needed for this validation. Consequently, context-environment gaps can remain undetected and lead to harmful outcomes. To evaluate whether agents identify and appropriately handle these gaps, we introduce [Benchmark Name], comprising [N] evaluation instances across [X] application scenarios covering three categories: (1) Requirement Underspecification, where requirements leave safety-critical choices unresolved; (2) Requirement Infeasibility, where requirements cannot be satisfied without violating safety constraints in the environment; and (3) Factual Inconsistency, where factual claims in context are incorrect or outdated relative to the current environment. For each scenario, we develop [Y] seed cases per category and expand them into evaluation instances. Evaluating [M] agents, we find [main quantitative result, specifying the metric and evaluation conditions]. Further analysis reveals [key failure pattern], showing [what this finding establishes about agent safety].

Blue brackets mark placeholders. X: application scenarios; Y: seed cases per scenario-category pair (3XY seeds in total); N: expanded evaluation instances; M: evaluated agents. Empirical findings remain to be filled in after evaluation.